

Author Contributions Checklist Form

This form documents the artifacts associated with the article (i.e., the data and code supporting the computational findings) and describes how to reproduce the findings.

Part 1: Data

☐ This paper **does not** involve analysis of external data (i.e., no data are used or the only data are generated by the authors via simulation in their code).

☒ I certify that the author(s) of the manuscript have legitimate access to and permission to use the data used in this manuscript.

Abstract

We analyze a dolphin-sighting dataset (n=936) from 'The World of Zero-Inflated Models,' utilizing covariates including depth, temperature, river distance, salinity, turbidity, and year. To address 75% zero-inflation, we fit hurdle Poisson, hurdle negative binomial (HNB), Poisson, and negative binomial (NB) regression models. We evaluate goodness-of-fit and covariate effects using Z-residual diagnostics and posterior predictive checking (PPC). Analysis-ready data (DolphinsThailandV2.csv) is included, and all simulation datasets are generated via the accompanying R code. The instructions to load/simulate datasets are given in the quarto (.qmd) files.

Availability

- ☒ Data **are** publicly available
☐ Data **cannot be made** publicly available

If the data are publicly available, see the *Publicly available data* section. Otherwise, see the *Non-publicly available data* section, below.

Publicly available data

☒ Data are available online at:

- Figshare (anonymized for review): <https://figshare.com/s/00ac6b406e451c76e8e1>
- Original data source: <https://www.highstat.com/index.php?view=article&id=10:the-world-of-zero-inflated-models&catid=18>

☒ Data are available as part of the paper's supplementary material.

A small-size **reproducibility-materials.zip** is submitted in the journal submission portal. This archive contains the R and Quarto files, along with the real dataset used to reproduce the empirical experiments.

☐ Data are publicly available by request, following the process described here:

☐ Data are or will be made available through some other mechanism, described here:

Non-publicly available data

Discussion of lack of publicly available data:

Description

File format(s)

☒ CSV or other plain text:

The file `DolphinsThailandV2.csv` is located in the `inst/extdata` directory of the `reproducibility-materials.zip` archive.

☐ Software-specific binary format (.Rda, Python pickle, etc.):

☐ Standardized binary format (e.g., netCDF, HDF5, etc.):

☐ Other (described here):

Data dictionary

☐ Provided by the authors in the following file(s):

☐ Data file(s) is (are) self-describing (e.g., netCDF files)

☐ Available at the following URL:

Additional information (optional)

NA

Part 2: Code

Abstract

- The supplementary code is provided as an R package, Zresidual, submitted as reproducibility-materials.zip. This package includes all R source files and Quarto-based vignette sources.
- The files available at <https://figshare.com/s/00ac6b406e451c76e8e1> include all contents of reproducibility-materials.zip, supplemented by pre-compiled HTML documentation generated via pkgdown. The documentation entry point is docs/index.html, which functions as a website.

Description

Code format(s)

☒ Script files

☒ R ☐ Python ☐ Matlab

☐ Other:

☒ Package

☒ R ☐ Python ☐ MATLAB toolbox

☐ Other:

☒ Reproducible report

☐ R Markdown ☐ Jupyter notebook

☒ Other: Quarto (.qmd)

☐ Shell script

☐ Other (described here):

Supporting software requirements

Version of primary software used

R version 4.5.1

Libraries and dependencies used by the code

Zresidual package is the accompanying software for this paper. Other R packages include brms, rstan, survival, ggplot2, dplyr, tidyr, tibble, loo, MASS, VGAM, actuar, matrixStats, distrib

utions3, gt, gifski, and knitr. Exact packages and their versions are recorded in the file `renv.lock` and stated explicitly in the vignette quarto files.

Supporting system/hardware requirements (optional)

N/A

Parallelization used

- ☒ No parallel code used
 - ☐ Multi-core parallelization on a single machine/node
 - ☐ Multi-machine/multi-node parallelization
- Number of nodes and cores used:

License

- ☒ MIT License (default)
- ☐ BSD
- ☐ GPL v3.0
- ☐ Creative Commons
- ☐ Other (described here):

Additional information (optional)

Part 3: Reproducibility workflow

Scope

The provided workflow reproduces:

- ☐ Any numbers provided in text in the paper
- ☒ The computational method(s) presented in the paper (i.e., code is provided that implements the method(s))
- ☐ All tables and figures in the paper
- ☒ Selected tables and figures in the paper, as explained and justified here:

This workflow reproduces the computational methods, simulation-based demonstrations, and empirical examples detailed in Sections 3.2, 3.3, S1.1, and S1.2 of the manuscript. The tables and figures presented in these reproducible vignettes are regenerated from the supplied R and Quarto source files, with pre-compiled HTML outputs provided. Please note that certain large cache objects for replicated studies are excluded and will be regenerated locally upon execution. Additionally, code for large-scale simulations is omitted, as these represent repeated executions of the demonstration files for a single dataset.

Workflow details

Format(s)

- ☐ Single master code file
- ☐ Wrapper (shell) script(s)
- ☐ Self-contained R Markdown file, Jupyter notebook, or other literate programming approach
- ☒ Text file (e.g., a readme-style file) that documents workflow
- ☐ Makefile
- ☒ Other (more detail in 'Instructions' below)

Instructions

All workflow files are provided as Quarto (.qmd) documents, available via:

- The Figshare archive: <https://figshare.com/s/00ac6b406e451c76e8e1>
- The reproducibility-materials.zip archive submitted to the journal portal.

The Quarto source files and complete reproduction materials are located in the vignettes directory, while pre-compiled HTML documentation is accessible via docs/index.html.

To reproduce the analyses, open a .qmd file in RStudio, VS Code, or Positron and render the document. Alternatively, you may open the entire directory as a project or workspace in your preferred IDE, or copy the embedded R code from the HTML files directly into an R console. If

using the provided R project, ensure all dependencies are installed via `renv::restore()` before rendering.

Expected run-time

Approximate time needed to reproduce the analyses on a standard desktop machine:

- ☐ <1 minute
- ☐ 1-10 minutes
- ☐ 10-60 minutes
- ☒ 1-8 hours
- ☐ >8 hours
- ☐ Not feasible to run on a desktop machine, as described here:

Additional documentation (optional)

Notes (optional)